Week : 8
الأسبوع #: من - إلى

Subject/Chemistry :الموضوع **Unit TitleStructure 1.4 – The mole : عنوان الوحدة** **Grade: 11**
(DP1)

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 S1.4.4–1.4.5 Lab :practical standard solutions + MgO empirical formula demo	**Learning target (DOK 2):** I can explain how to prepare a standard solution of known concentration, and use mass data from the magnesium oxide demonstration to determine and justify an empirical formula.	**Formative:** teacher observation of lab technique, lab notes and questioning during the demonstration. **Differentiation:** procedure checklist for Must; calculating concentration for Should; evaluating errors and uncertainty for Could. **UDL strategies:** 3.3 Guide processing with explicit steps, chunking, and interactive models. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	Video: Preparing a standard solution (Royal Society of Chemistry) Guide: Using a volumetric flask (University of Wisconsin–Madison) Lab procedure and demonstration notes are on the class Microsoft Teams channel.
Lesson 2 الحصة الثانية Summative exam	**Learning target (DOK 2):** I can apply mole concepts ($n = m/M$, $N = n \times N_A$, $c = n/V$ and empirical formulae) to solve and justify multi-step problems in a summative exam.	**Summative:** written exam. **Differentiation:** exam questions move from recall to multi-step problems. **UDL strategies:** 1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	Video: Molecular and empirical formulas (Khan Academy) Video: Formula from mass composition (Khan Academy) MCQ practice: Structure 1 practice questions, S1.1–S1.5 (iPassed) Revision materials are on the class Microsoft Teams channel.

Lesson 3 **الحصة الثالثة** **Summative exam (continued)**	**Learning target (DOK 2):** I can apply mole concepts (n = m/M, N = n × NA, c = n/V and empirical formulae) to solve and justify multi-step problems in a summative exam.	**Summative:** written exam (continued). **Differentiation:** exam questions move from recall to multi-step problems. **UDL strategies:** 1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	Video: Molecular and empirical formulas from percent composition (Khan Academy) Video: Formula from mass composition (Khan Academy) MCQ practice: Structure 1 practice questions, S1.1–S1.5 (iPassed) Students should bring a calculator and pens.
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Hello Parents, Please encourage your child to review and practice Science regularly each week using the resources available in their class Microsoft Teams - class material folder which is organized by week. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Science study guides will provide basic assessment information, topics, and expectations; however, they are intended as a guide only and will not contain detailed study materials, these are found in the class teams. Effective preparation for graded formatives and summatives requires students to actively engage with the learning materials provided in their class teams. Lesson 1 lab safety: goggles, lab coats and closed shoes are required. Exam (Lessons 2–3): calculators and pens only; no phones. Thank you for partnering with us to support your child's learning and success. Ms. Stacey (Science Coordinator)		

Week # الأسبوع#

Technology Integration دمج التكنولوجيا	Digital balance; scientific calculators; digital IB data booklet; slides shared after each lesson.
Cross-curricular links الروابط بين المواد الدراسية	**Mathematics:** standard form, ratios and significant figures. **Physics:** SI units.
External Assessment Integration دمج التقييمات الخارجية	IB DP Chemistry Papers 1 and 2: data-booklet calculations and multi-step structured questions using the command terms calculate, determine, deduce and justify. Practical skills and uncertainty also support the Internal Assessment (IA).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking:** justifying answers and evaluating error. **Self-management:** exam preparation and time management. **Social:** teamwork and shared responsibility in the lab.
Standards المعايير	• **S1.4.4** The empirical formula gives the simplest ratio of atoms in a compound; the molecular formula gives the actual number of atoms in a molecule. • **S1.4.5** Molar concentration is determined by the amount of solute and the volume of solution.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Guiding question:** How do we quantify matter on the atomic scale? **Lesson 1:** How can we improve the accuracy of a standard solution, and how certain can we be of a formula deduced from mass measurements?
Success Criteria معايير النجاح **(Measurable)**	● **Must (DOK 1):** describe the steps to prepare a standard solution and state the empirical formula of magnesium oxide. ■ **Should (DOK 2):** calculate concentration using c = n/V and calculate an empirical formula from mass data. ▲ **Could (DOK 3):** solve multi-step problems, evaluate sources of error and justify each step. **Evidence:** lab notes, demonstration questioning and the summative exam.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Global Understanding التفاهم العالمي Heritage التراث
	المجال Area: Innovation and Infrastructure
	التطبيق Implementation: Students apply mole and concentration calculations to UAE contexts such as brine from desalination, and recognise the contributions of early Arab and Islamic scholars to chemistry.